

```
In [22]: pip install tensorflow numpy matplotlib scikit-learn
```

```
Requirement already satisfied: tensorflow in /usr/local/lib/python3.13/dist-packages (2.20.0)
Requirement already satisfied: numpy in /usr/local/lib/python3.13/dist-packages (2.1.3)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.13/dist-packages (3.10.0)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.13/dist-packages (1.6.1)
Requirement already satisfied: absl-py>=1.0.0 in /usr/local/lib/python3.13/dist-packages (from tensorflow) (1.4.0)
Requirement already satisfied: astunparse>=1.6.0 in /usr/local/lib/python3.13/dist-packages (from tensorflow) (1.6.3)
Requirement already satisfied: flatbuffers>=24.3.25 in /usr/local/lib/python3.13/dist-packages (from tensorflow) (25.12.19)
Requirement already satisfied: gast!=0.5.0,!0.5.1,!0.5.2,>=0.2.1 in /usr/local/lib/python3.13/dist-packages (from tensorflow) (0.7.0)
Requirement already satisfied: google_pasta>=0.1.1 in /usr/local/lib/python3.13/dist-packages (from tensorflow) (0.2.0)
Requirement already satisfied: libclang>=13.0.0 in /usr/local/lib/python3.13/dist-packages (from tensorflow) (18.1.1)
Requirement already satisfied: opt_einsum>=2.3.2 in /usr/local/lib/python3.13/dist-packages (from tensorflow) (3.4.0)
Requirement already satisfied: packaging in /usr/local/lib/python3.13/dist-packages (from tensorflow) (26.3)
Requirement already satisfied: protobuf>=5.28.0 in /usr/local/lib/python3.13/dist-packages (from tensorflow) (5.29.6)
Requirement already satisfied: requests<3,>=2.21.0 in /usr/local/lib/python3.13/dist-packages (from tensorflow) (2.32.4)
Requirement already satisfied: setuptools in /usr/local/lib/python3.13/dist-packages (from tensorflow) (80.10.2)
Requirement already satisfied: six>=1.12.0 in /usr/local/lib/python3.13/dist-packages (from tensorflow) (1.17.0)
Requirement already satisfied: termcolor>=1.1.0 in /usr/local/lib/python3.13/dist-packages (from tensorflow) (3.3.0)
Requirement already satisfied: typing_extensions>=3.6.6 in /usr/local/lib/python3.13/dist-packages (from tensorflow) (4.16.0)
Requirement already satisfied: wrapt>=1.11.0 in /usr/local/lib/python3.13/dist-packages (from tensorflow) (2.4.1)
Requirement already satisfied: grpcio<2.0,>=1.24.3 in /usr/local/lib/python3.13/dist-packages (from tensorflow) (1.84.0)
Requirement already satisfied: tensorboard~=2.20.0 in /usr/local/lib/python3.13/dist-packages (from tensorflow) (2.20.0)
Requirement already satisfied: keras>=3.10.0 in /usr/local/lib/python3.13/dist-packages (from tensorflow) (3.13.2)
Requirement already satisfied: h5py>=3.11.0 in /usr/local/lib/python3.13/dist-packages (from tensorflow) (3.16.0)
Requirement already satisfied: ml_dtypes<1.0.0,>=0.5.1 in /usr/local/lib/python3.13/dist-packages (from tensorflow) (0.6.0)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (1.4.0)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (4.65.0)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (1.5.1)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.13/dist-packages (from scikit-learn) (1.16.3)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.13/dist-packages (from scikit-learn) (1.6.0)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.13/dist-packages (from scikit-learn) (3.7.0)
Requirement already satisfied: wheel<1.0,>=0.23.0 in /usr/local/lib/python3.13/dist-packages (from astunparse>=1.6.0->tensorflow) (0.48.0)
Requirement already satisfied: cloudpickle>=3.0 in /usr/local/lib/python3.13/dist-packages (from joblib>=1.2.0->scikit-learn) (3.1.2)
Requirement already satisfied: rich in /usr/local/lib/python3.13/dist-packages (from keras>=3.10.0->tensorflow) (13.9.4)
Requirement already satisfied: namex in /usr/local/lib/python3.13/dist-packages (from keras>=3.10.0->tensorflow) (0.1.0)
Requirement already satisfied: optree in /usr/local/lib/python3.13/dist-packages (from keras>=3.10.0->tensorflow) (0.20.0)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.13/dist-packages (from requests<3,>=2.21.0->tensorflow) (3.4.9)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.13/dist-packages (from requests<3,>=2.21.0->tensorflow) (3.19)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.13/dist-packages (from requests<3,>=2.21.0->tensorflow) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.13/dist-packages (from requests<3,>=2.21.0->tensorflow) (2026.7.22)
Requirement already satisfied: markdown>=2.6.8 in /usr/local/lib/python3.13/dist-packages (from tensorboard~=2.20.0->tensorflow) (3.10.3)
Requirement already satisfied: tensorboard-data-server<0.8.0,>=0.7.0 in /usr/local/lib/python3.13/dist-packages (from tensorboard~=2.20.0->tensorflow) (0.7.2)
Requirement already satisfied: werkzeug>=1.0.1 in /usr/local/lib/python3.13/dist-packages (from tensorboard~=2.20.0->tensorflow) (3.1.8)
Requirement already satisfied: markupsafe>=2.1.1 in /usr/local/lib/python3.13/dist-packages (from werkzeug>=1.0.1->tensorboard~=2.20.0->tensorflow) (3.0.3)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.13/dist-packages (from rich->keras>=3.10.0->tensorflow) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.13/dist-packages (from rich->keras>=3.10.0->tensorflow) (2.21.0)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.13/dist-packages (from markdown-it-py>=2.2.0->rich->keras>=3.10.0->tensorflow) (0.1.2)
```

```
In [23]: !pip install tensorflow numpy matplotlib scikit-learn
```

Requirement already satisfied: tensorflow in /usr/local/lib/python3.13/dist-packages (2.20.0)
Requirement already satisfied: numpy in /usr/local/lib/python3.13/dist-packages (2.1.3)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.13/dist-packages (3.10.0)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.13/dist-packages (1.6.1)
Requirement already satisfied: absl-py>=1.0.0 in /usr/local/lib/python3.13/dist-packages (from tensorflow) (1.4.0)
Requirement already satisfied: astunparse>=1.6.0 in /usr/local/lib/python3.13/dist-packages (from tensorflow) (1.6.3)
Requirement already satisfied: flatbuffers>=24.3.25 in /usr/local/lib/python3.13/dist-packages (from tensorflow) (25.12.19)
Requirement already satisfied: gast!=0.5.0,!0.5.1,!0.5.2,>=0.2.1 in /usr/local/lib/python3.13/dist-packages (from tensorflow) (0.7.0)
Requirement already satisfied: google_pasta>=0.1.1 in /usr/local/lib/python3.13/dist-packages (from tensorflow) (0.2.0)
Requirement already satisfied: libclang>=13.0.0 in /usr/local/lib/python3.13/dist-packages (from tensorflow) (18.1.1)
Requirement already satisfied: opt_einsum>=2.3.2 in /usr/local/lib/python3.13/dist-packages (from tensorflow) (3.4.0)
Requirement already satisfied: packaging in /usr/local/lib/python3.13/dist-packages (from tensorflow) (26.3)
Requirement already satisfied: protobuf>=5.28.0 in /usr/local/lib/python3.13/dist-packages (from tensorflow) (5.29.6)
Requirement already satisfied: requests<3,>=2.21.0 in /usr/local/lib/python3.13/dist-packages (from tensorflow) (2.32.4)
Requirement already satisfied: setuptools in /usr/local/lib/python3.13/dist-packages (from tensorflow) (80.10.2)
Requirement already satisfied: six>=1.12.0 in /usr/local/lib/python3.13/dist-packages (from tensorflow) (1.17.0)
Requirement already satisfied: termcolor>=1.1.0 in /usr/local/lib/python3.13/dist-packages (from tensorflow) (3.3.0)
Requirement already satisfied: typing_extensions>=3.6.6 in /usr/local/lib/python3.13/dist-packages (from tensorflow) (4.16.0)
Requirement already satisfied: wrapt>=1.11.0 in /usr/local/lib/python3.13/dist-packages (from tensorflow) (2.4.1)
Requirement already satisfied: grpcio<2.0,>=1.24.3 in /usr/local/lib/python3.13/dist-packages (from tensorflow) (1.84.0)
Requirement already satisfied: tensorboard~=2.20.0 in /usr/local/lib/python3.13/dist-packages (from tensorflow) (2.20.0)
Requirement already satisfied: keras>=3.10.0 in /usr/local/lib/python3.13/dist-packages (from tensorflow) (3.13.2)
Requirement already satisfied: h5py>=3.11.0 in /usr/local/lib/python3.13/dist-packages (from tensorflow) (3.16.0)
Requirement already satisfied: ml_dtypes<1.0.0,>=0.5.1 in /usr/local/lib/python3.13/dist-packages (from tensorflow) (0.6.0)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (1.4.0)
Requirement already satisfied: cycycler>=0.10 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (4.65.0)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (1.5.1)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.13/dist-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.13/dist-packages (from scikit-learn) (1.16.3)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.13/dist-packages (from scikit-learn) (1.6.0)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.13/dist-packages (from scikit-learn) (3.7.0)
Requirement already satisfied: wheel<1.0,>=0.23.0 in /usr/local/lib/python3.13/dist-packages (from astunparse>=1.6.0->tensorflow) (0.48.0)
Requirement already satisfied: cloudpickle>=3.0 in /usr/local/lib/python3.13/dist-packages (from joblib>=1.2.0->scikit-learn) (3.1.2)
Requirement already satisfied: rich in /usr/local/lib/python3.13/dist-packages (from keras>=3.10.0->tensorflow) (13.9.4)
Requirement already satisfied: namex in /usr/local/lib/python3.13/dist-packages (from keras>=3.10.0->tensorflow) (0.1.0)
Requirement already satisfied: optree in /usr/local/lib/python3.13/dist-packages (from keras>=3.10.0->tensorflow) (0.20.0)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.13/dist-packages (from requests<3,>=2.21.0->tensorflow) (3.4.9)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.13/dist-packages (from requests<3,>=2.21.0->tensorflow) (3.19)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.13/dist-packages (from requests<3,>=2.21.0->tensorflow) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.13/dist-packages (from requests<3,>=2.21.0->tensorflow) (2026.7.22)
Requirement already satisfied: markdown>=2.6.8 in /usr/local/lib/python3.13/dist-packages (from tensorboard~=2.20.0->tensorflow) (3.10.3)
Requirement already satisfied: tensorboard-data-server<0.8.0,>=0.7.0 in /usr/local/lib/python3.13/dist-packages (from tensorboard~=2.20.0->tensorflow) (0.7.2)
Requirement already satisfied: werkzeug>=1.0.1 in /usr/local/lib/python3.13/dist-packages (from tensorboard~=2.20.0->tensorflow) (3.1.8)
Requirement already satisfied: markupsafe>=2.1.1 in /usr/local/lib/python3.13/dist-packages (from werkzeug>=1.0.1->tensorboard~=2.20.0->tensorflow) (3.0.3)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.13/dist-packages (from rich->keras>=3.10.0->tensorflow) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.13/dist-packages (from rich->keras>=3.10.0->tensorflow) (2.21.0)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.13/dist-packages (from markdown-it-py>=2.2.0->rich->keras>=3.10.0->tensorflow) (0.1.2)

```
In [24]: import numpy as np
import matplotlib.pyplot as plt

import tensorflow as tf
from tensorflow.keras import layers, models

from sklearn.metrics import (
    confusion_matrix,
    classification_report
)
```

```
In [25]: (x_train, y_train), (x_test, y_test) = tf.keras.datasets.mnist.load_data()

print("Training images:", x_train.shape)
print("Training labels:", y_train.shape)

print("Testing images:", x_test.shape)
print("Testing labels:", y_test.shape)
```

```
Training images: (60000, 28, 28)
Training labels: (60000,)
Testing images: (10000, 28, 28)
Testing labels: (10000,)
```

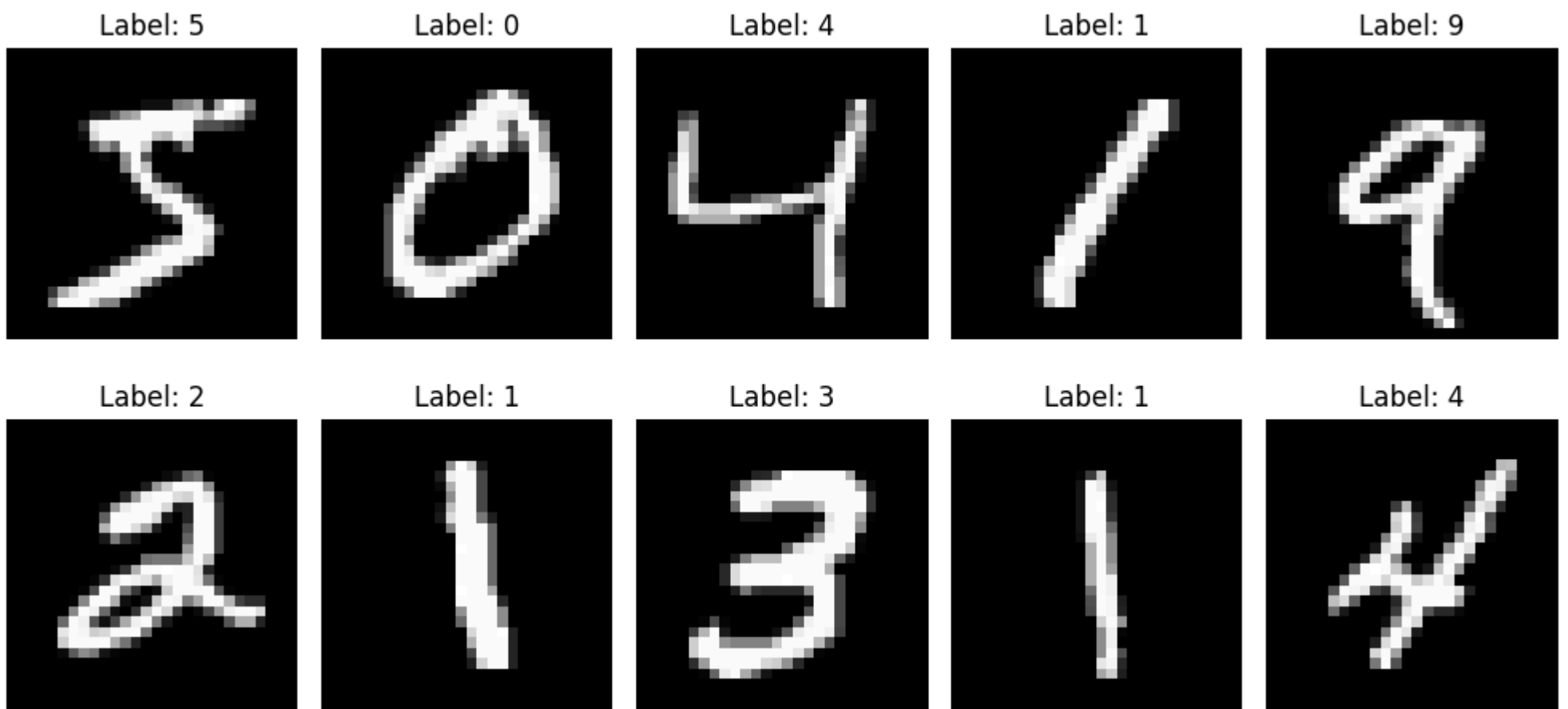
```
In [26]: plt.figure(figsize=(10, 5))
```

```

for i in range(10):
    plt.subplot(2, 5, i + 1)
    plt.imshow(x_train[i], cmap="gray")
    plt.title(f"Label: {y_train[i]}")
    plt.axis("off")

plt.tight_layout()
plt.show()

```



8. Step 4 – Image Preprocessing

The original pixel values range from:

0 → 255

We normalize them to:

0 → 1

using:

$$x_{normalized} = \frac{x}{255}$$

Code

```

In [27]: x_train = x_train.astype("float32") / 255.0
x_test = x_test.astype("float32") / 255.0

```

13. Step 5: Reshape the Images

CNN expects image data in the format:

samples × height × width × channels

Since MNIST is grayscale, it has one channel

```

In [28]: x_train = x_train.reshape(-1, 28, 28, 1)
x_test = x_test.reshape(-1, 28, 28, 1)

```

```

print(x_train.shape)
print(x_test.shape)

```

```

(60000, 28, 28, 1)
(10000, 28, 28, 1)

```

```

In [29]: model = models.Sequential([

    layers.Conv2D(
        32,
        (3, 3),
        activation="relu",
        input_shape=(28, 28, 1)
    ),

    layers.MaxPooling2D((2, 2)),

    layers.Conv2D(
        64,
        (3, 3),

```

```
        activation="relu"
    ),

    layers.MaxPooling2D((2, 2)),

    layers.Flatten(),

    layers.Dense(
        128,
        activation="relu"
    ),

    layers.Dropout(0.3),

    layers.Dense(
        10,
        activation="softmax"
    )
])
```

/usr/local/lib/python3.13/dist-packages/keras/src/layers/convolutional/base_conv.py:113: UserWarning: Do not pass an `input\_shape`/`input\_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.
super().__init__(activity_regularizer=activity_regularizer, **kwargs)

15. CNN Architecture

Our model can be represented as:

Input $28 \times 28 \times 1$ $\downarrow$ Conv2D 32 filters 3×3 $\downarrow$ ReLU $\downarrow$ MaxPooling 2×2 $\downarrow$ Conv2D 64 filters 3×3 $\downarrow$ ReLU $\downarrow$ MaxPooling 2×2 $\downarrow$ Flatten $\downarrow$ Dense 128 neurons $\downarrow$ Dropout $\downarrow$ Dense 10 neurons $\downarrow$ Softmax $\downarrow$ Digit 0–9

16. Step 7: Display Model Summary

In [30]: `model.summary()`

Model: "sequential_1"

Layer (type)	Output Shape	Param #
conv2d_2 (Conv2D)	(None, 26, 26, 32)	320
max_pooling2d_2 (MaxPooling2D)	(None, 13, 13, 32)	0
conv2d_3 (Conv2D)	(None, 11, 11, 64)	18,496
max_pooling2d_3 (MaxPooling2D)	(None, 5, 5, 64)	0
flatten_1 (Flatten)	(None, 1600)	0
dense_2 (Dense)	(None, 128)	204,928
dropout_1 (Dropout)	(None, 128)	0
dense_3 (Dense)	(None, 10)	1,290

Total params: 225,034 (879.04 KB)
Trainable params: 225,034 (879.04 KB)
Non-trainable params: 0 (0.00 B)

In [31]: `model.compile(
 optimizer="adam",
 loss="sparse_categorical_crossentropy",
 metrics=["accuracy"]
)`

Explanation

Optimizer: Adam

It updates the model weights during training.

Loss function: Sparse categorical cross-entropy

It is suitable when class labels are represented as integers such as:

0, 1, 2, ..., 9

Metric: Accuracy

Measures the percentage of correctly classified images.

18. Step 9: Train the CNN

In [32]: `history = model.fit(
 x_train,
 y_train,
 epochs=5,`

```
batch_size=64,  
validation_split=0.1  
)
```

Epoch 1/5

844/844 ————— **59s** 68ms/step - accuracy: 0.9345 - loss: 0.2123 - val_accuracy: 0.9865 - val_loss: 0.0483

Epoch 2/5

844/844 ————— **41s** 48ms/step - accuracy: 0.9802 - loss: 0.0654 - val_accuracy: 0.9887 - val_loss: 0.0414

Epoch 3/5

844/844 ————— **42s** 50ms/step - accuracy: 0.9861 - loss: 0.0455 - val_accuracy: 0.9903 - val_loss: 0.0327

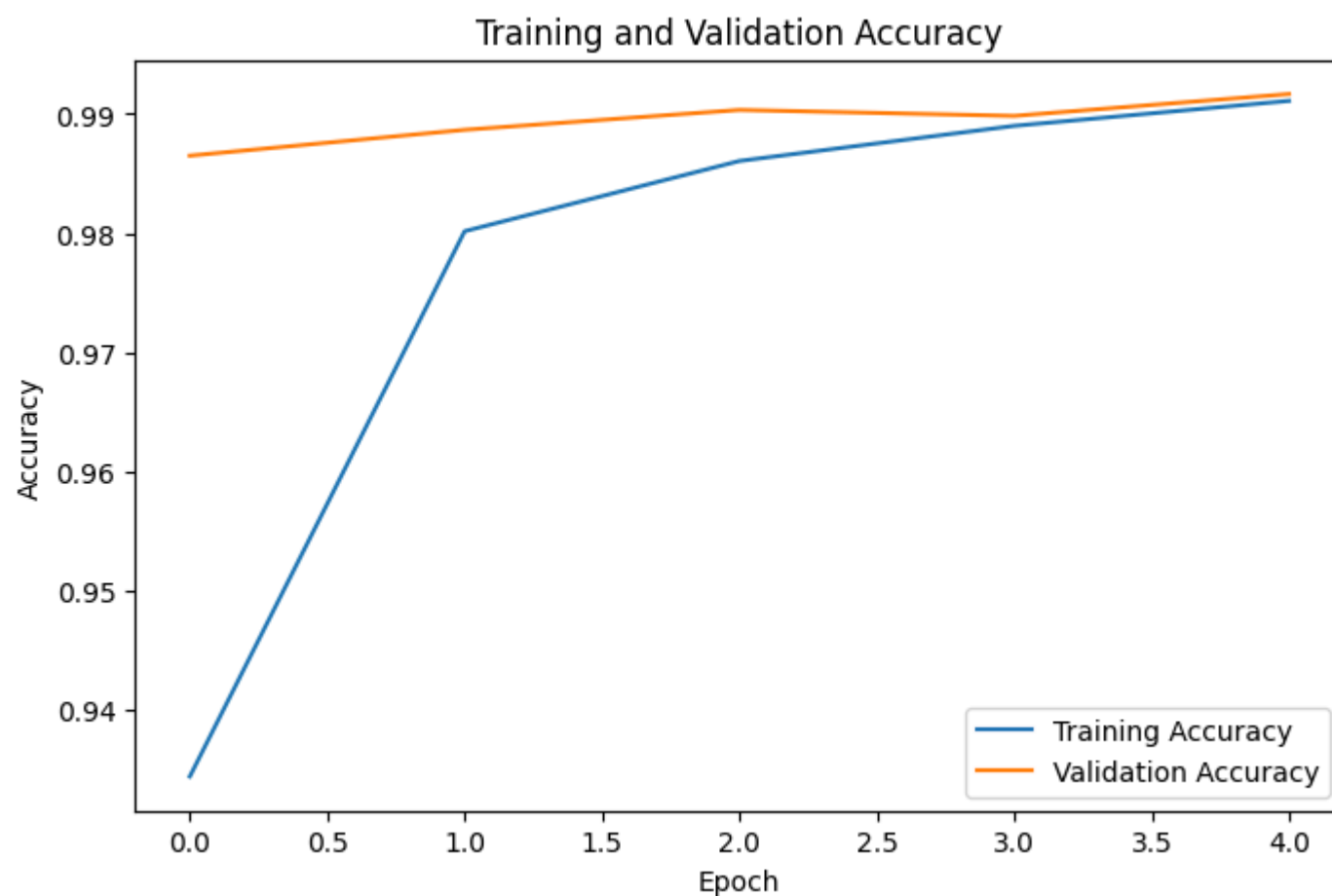
Epoch 4/5

844/844 ————— **42s** 49ms/step - accuracy: 0.9890 - loss: 0.0354 - val_accuracy: 0.9898 - val_loss: 0.0373

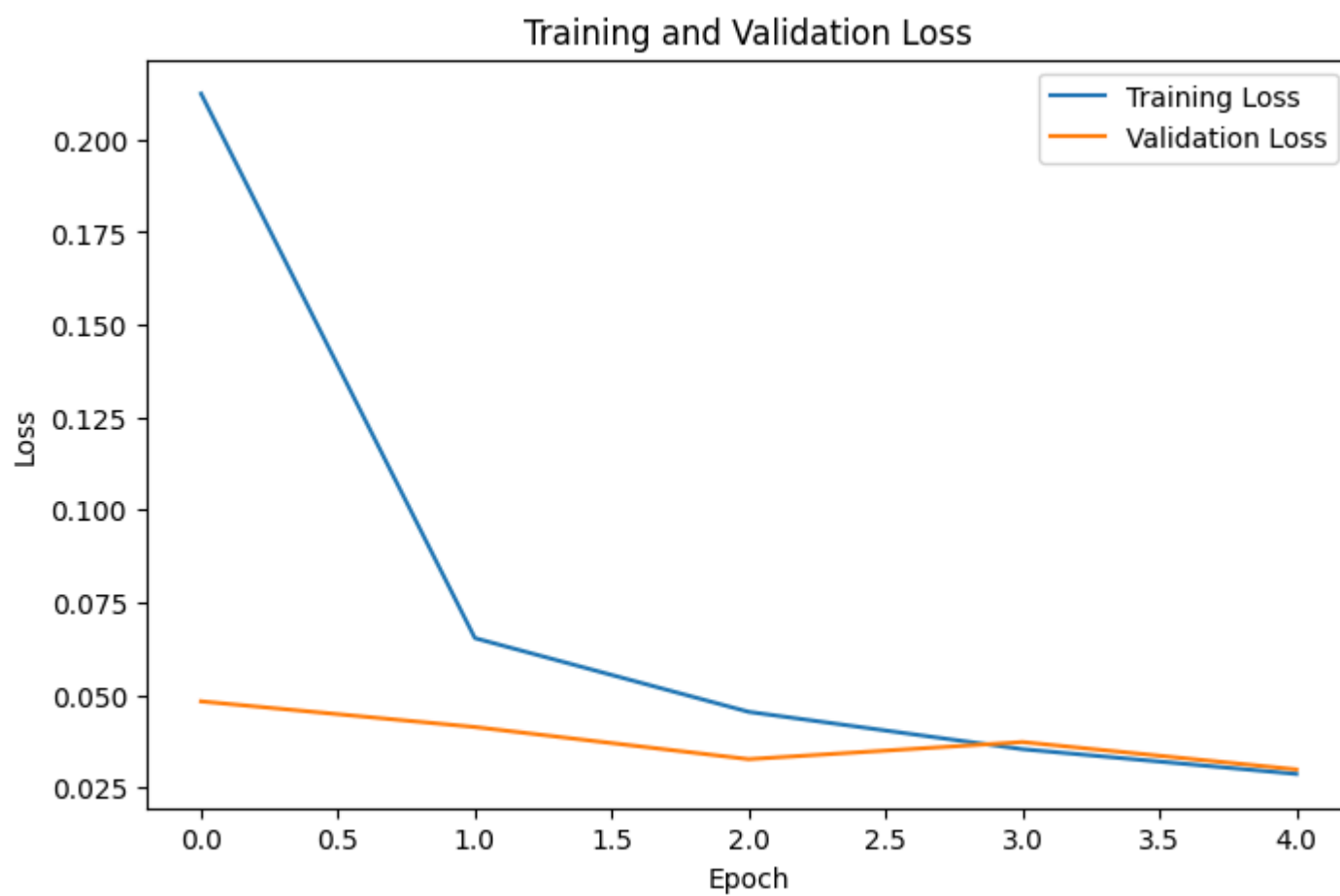
Epoch 5/5

844/844 ————— **41s** 48ms/step - accuracy: 0.9911 - loss: 0.0287 - val_accuracy: 0.9917 - val_loss: 0.0299

```
In [33]: plt.figure(figsize=(8, 5))  
  
plt.plot(  
    history.history["accuracy"],  
    label="Training Accuracy"  
)  
  
plt.plot(  
    history.history["val_accuracy"],  
    label="Validation Accuracy"  
)  
  
plt.xlabel("Epoch")  
plt.ylabel("Accuracy")  
plt.title("Training and Validation Accuracy")  
plt.legend()  
plt.show()
```



```
In [34]: plt.figure(figsize=(8, 5))  
  
plt.plot(  
    history.history["loss"],  
    label="Training Loss"  
)  
  
plt.plot(  
    history.history["val_loss"],  
    label="Validation Loss"  
)  
  
plt.xlabel("Epoch")  
plt.ylabel("Loss")  
plt.title("Training and Validation Loss")  
plt.legend()  
plt.show()
```



```
In [35]: test_loss, test_accuracy = model.evaluate(
        x_test,
        y_test,
        verbose=0
    )

    print("Test Loss:", test_loss)
    print("Test Accuracy:", test_accuracy)
```

Test Loss: 0.024801786988973618
 Test Accuracy: 0.991100013256073

```
In [36]: predictions = model.predict(x_test)

    predicted_labels = np.argmax(
        predictions,
        axis=1
    )

    print("Predicted:", predicted_labels[:10])
    print("Actual   :", y_test[:10])
```

313/313 ————— **3s** 9ms/step
 Predicted: [7 2 1 0 4 1 4 9 5 9]
 Actual : [7 2 1 0 4 1 4 9 5 9]

```
In [37]: plt.figure(figsize=(10, 5))

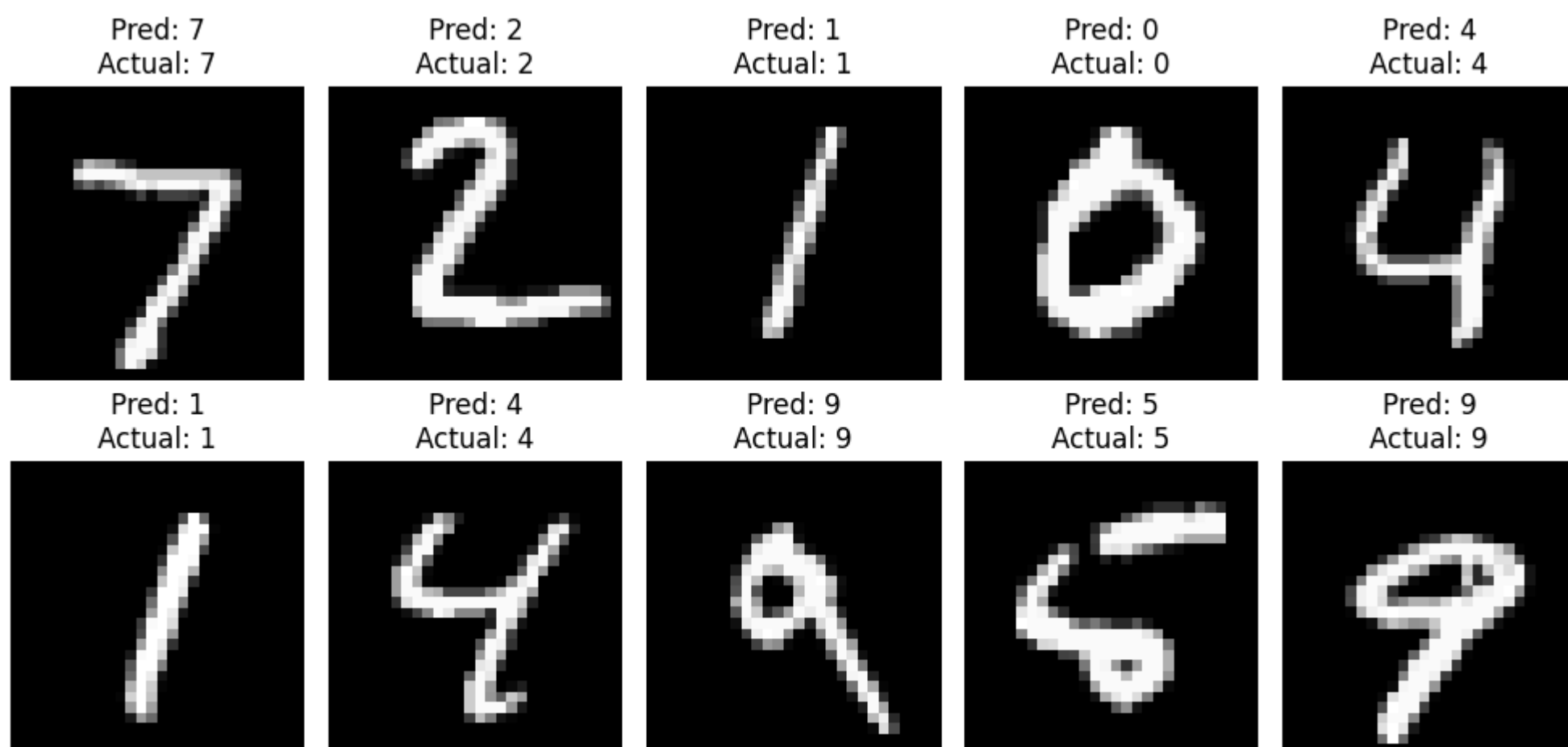
    for i in range(10):
        plt.subplot(2, 5, i + 1)

        plt.imshow(
            x_test[i].reshape(28, 28),
            cmap="gray"
        )

        plt.title(
            f"Pred: {predicted_labels[i]}\n"
            f"Actual: {y_test[i]}"
        )

        plt.axis("off")

    plt.tight_layout()
    plt.show()
```



```
In [38]: cm = confusion_matrix(
          y_test,
          predicted_labels
        )

print(cm)
```

```
[ [ 975    0    1    0    1    1    0    1    1    0]
  [   0 1128    2    1    0    1    0    2    1    0]
  [   0    0 1026    2    1    0    0    2    1    0]
  [   0    0    1 1002    0    6    0    0    1    0]
  [   0    0    1    0 979    0    0    0    0    2]
  [   1    0    0    5    0 883    1    1    0    1]
  [   4    3    0    0    2    7 941    0    1    0]
  [   0    0    3    2    0    0    0 1019    1    3]
  [   3    0    2    1    0    3    0    2 961    2]
  [   0    0    0    1    3    5    0    1    2 997]]
```

```
In [39]: plt.figure(figsize=(8, 6))

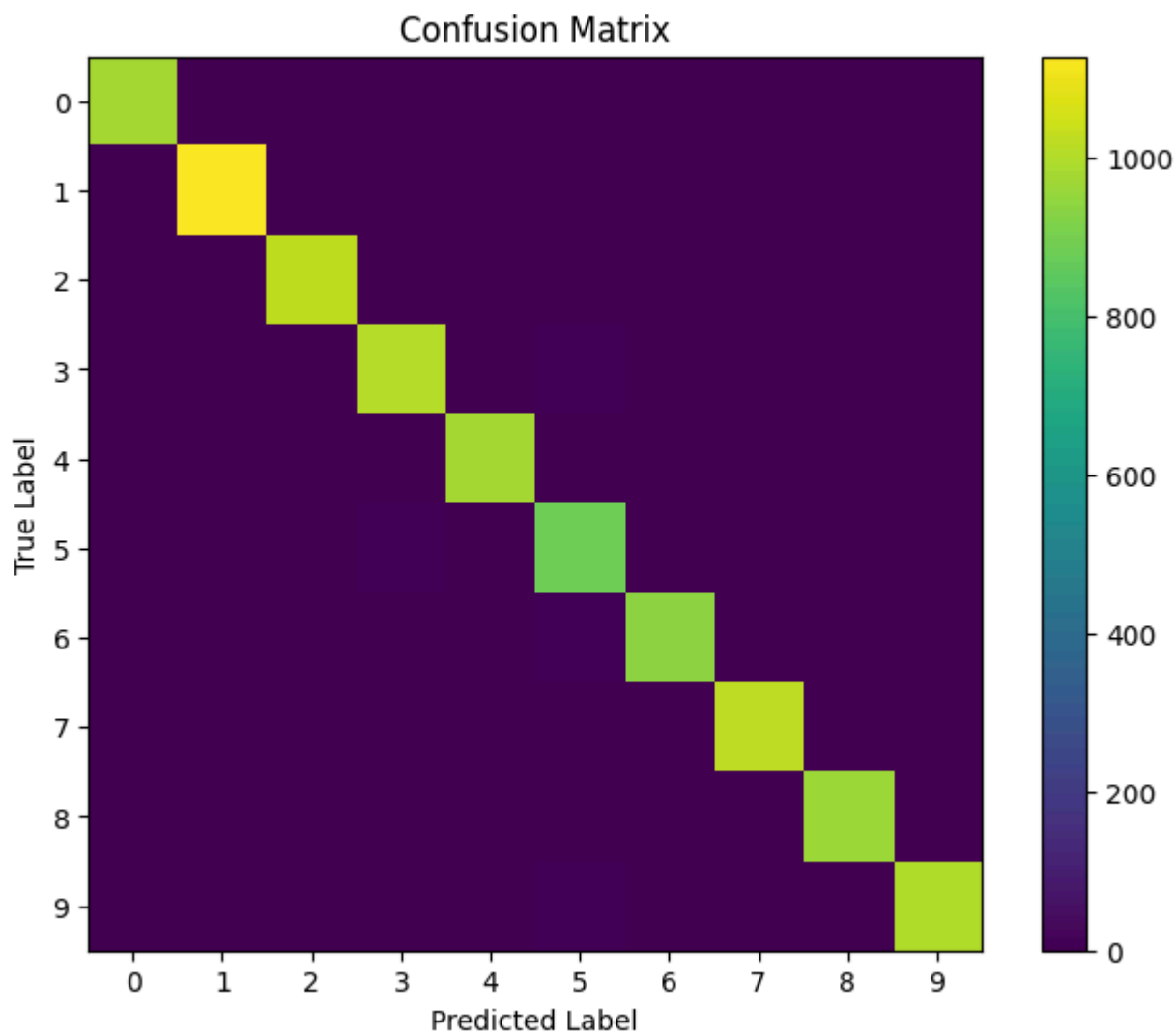
plt.imshow(cm)

plt.title("Confusion Matrix")
plt.xlabel("Predicted Label")
plt.ylabel("True Label")

plt.colorbar()

plt.xticks(range(10))
plt.yticks(range(10))

plt.show()
```



```
In [40]: print(
    classification_report(
        y_test,
        predicted_labels
    )
)
```

	precision	recall	f1-score	support
0	0.99	0.99	0.99	980
1	1.00	0.99	1.00	1135
2	0.99	0.99	0.99	1032
3	0.99	0.99	0.99	1010
4	0.99	1.00	0.99	982
5	0.97	0.99	0.98	892
6	1.00	0.98	0.99	958
7	0.99	0.99	0.99	1028
8	0.99	0.99	0.99	974
9	0.99	0.99	0.99	1009
accuracy			0.99	10000
macro avg	0.99	0.99	0.99	10000
weighted avg	0.99	0.99	0.99	10000

```
In [41]: wrong = np.where(
    predicted_labels != y_test
)[0]

print("Number of misclassified images:", len(wrong))
```

Number of misclassified images: 89

```
In [42]: plt.figure(figsize=(10, 6))

for i in range(min(10, len(wrong))):
    index = wrong[i]

    plt.subplot(2, 5, i + 1)

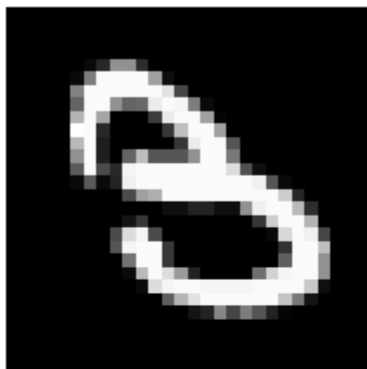
    plt.imshow(
        x_test[index].reshape(28, 28),
        cmap="gray"
    )

    plt.title(
        f"Pred: {predicted_labels[index]}, "
        f"Actual: {y_test[index]}"
    )

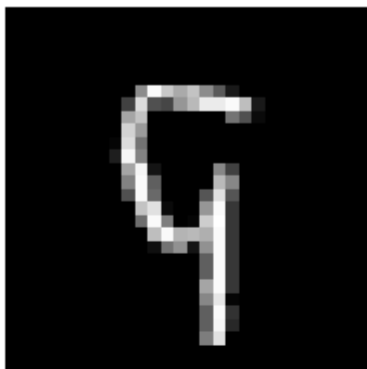
    plt.axis("off")

plt.tight_layout()
plt.show()
```

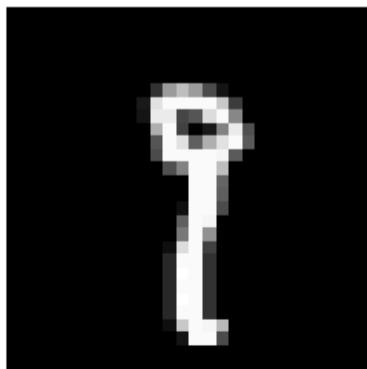
Pred: 5, Actual: 3



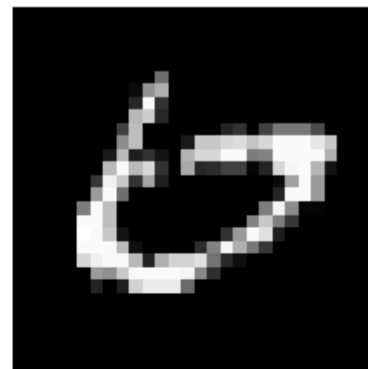
Pred: 5, Actual: 9



Pred: 8, Actual: 9



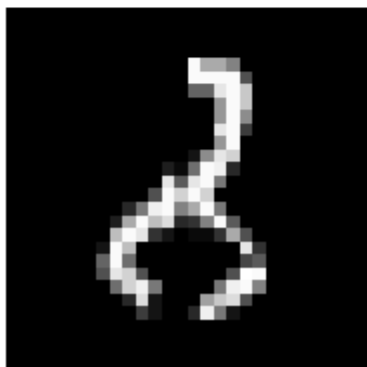
Pred: 0, Actual: 6



Pred: 5, Actual: 3



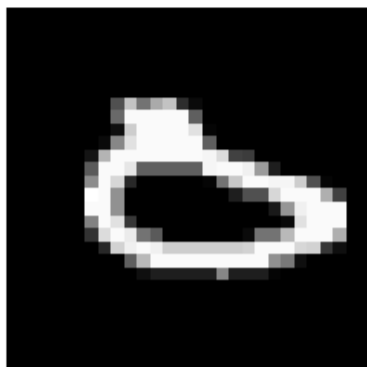
Pred: 2, Actual: 8



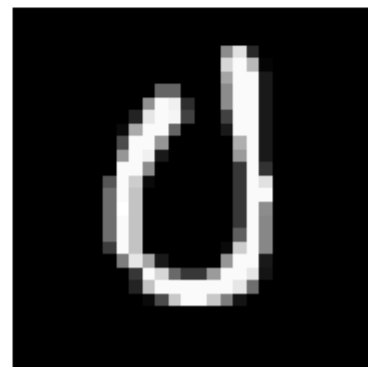
Pred: 4, Actual: 6



Pred: 5, Actual: 0



Pred: 4, Actual: 0



Pred: 5, Actual: 3

